Supplementary Information for Branch-specific detection of phylogenetic signal with SatuTe

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Supplementary Results

Baseline methodological controls

The baseline simulations remain necessary controls for the extension. They establish the gold-standard power curve, expose inflation from selecting and testing a topology on the same alignment, show the benefit and possible conservatism of taxon-pair Bonferroni correction, and document the distinct effect of substitution-model misspecification. The newly generated JC, GTR, protein and misspecification panels below reproduce these methodological comparisons using the current implementation and verified publication grids; no artwork from the earlier paper is reused.

Topology-selection and bootstrap controls

Model	Sites	Branch t	Counterfactual	Refined	Difference
$+I+G4$	100	50	0.100	0.115	0.015
	500	40	0.342	0.384	0.042
	1,000	45	0.406	0.456	0.050
	5,000	60	0.482	0.584	0.102
	10,000	60	0.718	0.835	0.116
$+I+R4$	100	10	0.558	0.573	0.015
	500	20	0.405	0.450	0.044
	1,000	22.5	0.449	0.522	0.073
	5,000	30	0.501	0.605	0.104
	10,000	35	0.402	0.565	0.163

Table S1: **Largest calibrated power gain at each alignment length in the invariant-mixture grid.** Differences are refined minus counterfactual; both formulas evaluate every alignment.

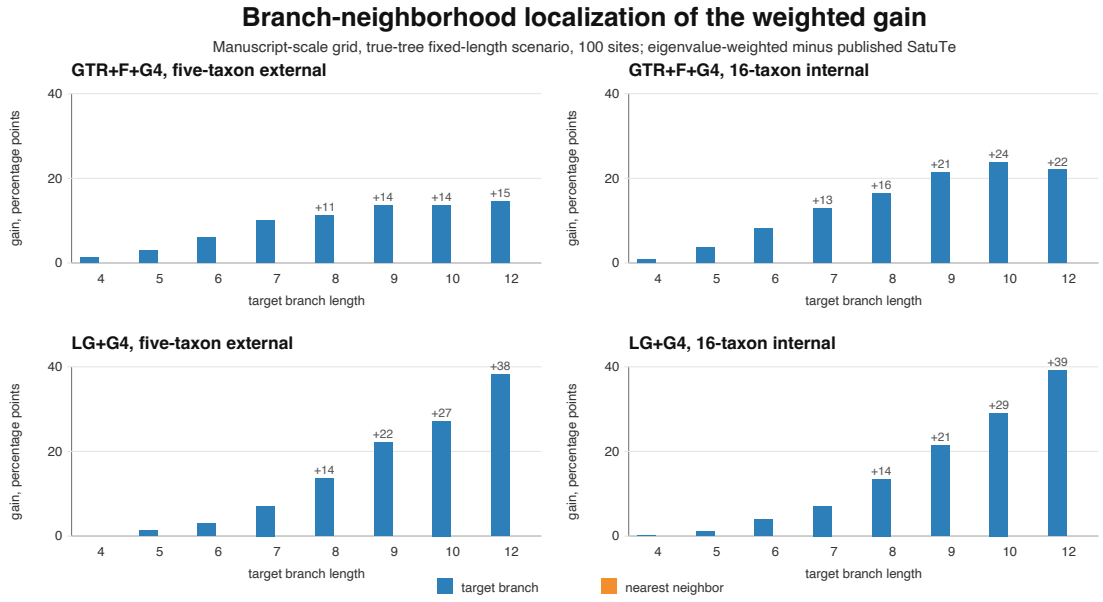


Figure S1: **Branch-neighbourhood localization.** Eigenvalue-weighted-minus-published-SatuTe changes in the fraction of phylogenetically informative alignments for the tested branch and adjacent branches. The change is concentrated on the deliberately lengthened focal branch.

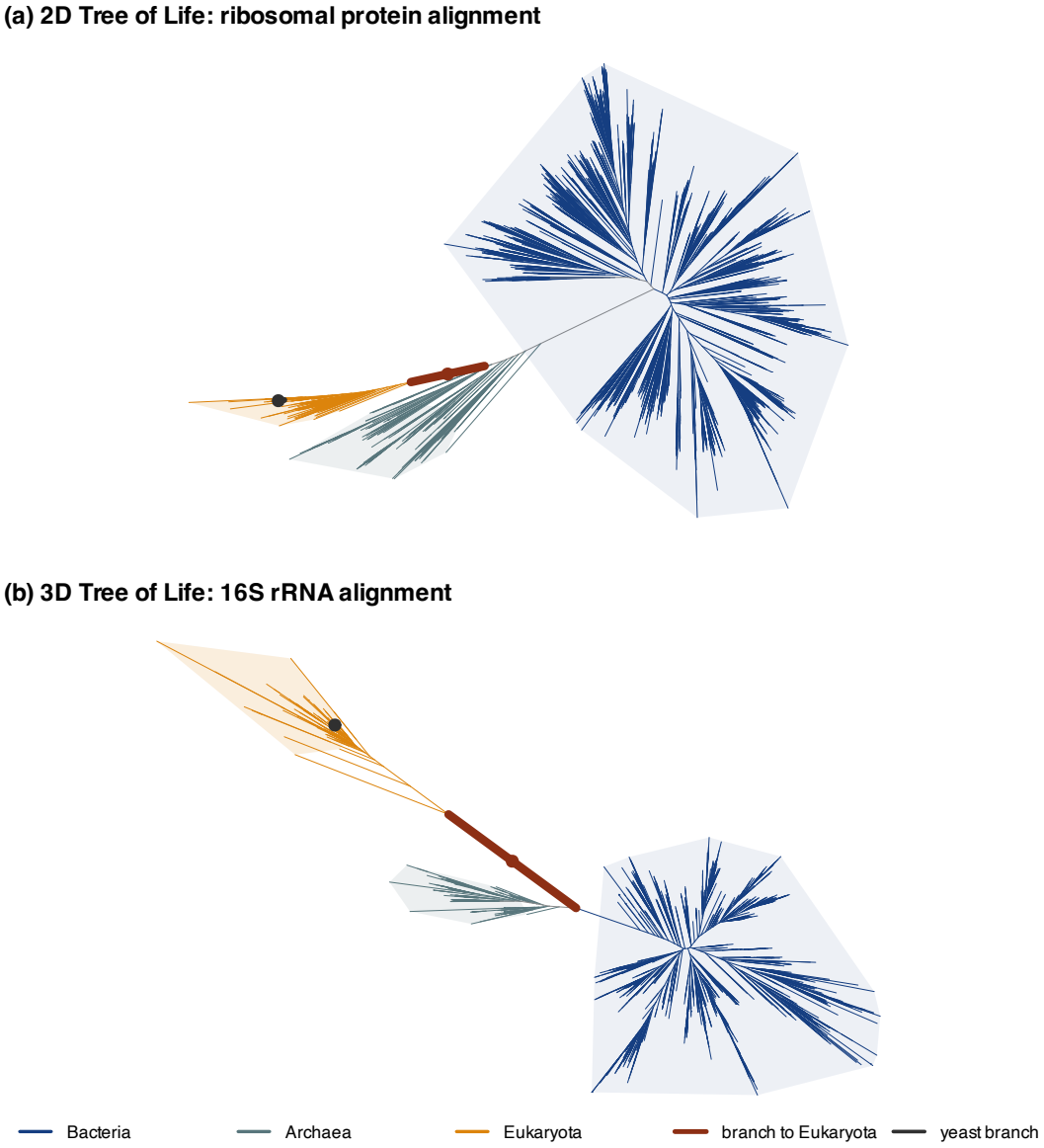


Figure S2: **Tree-of-Life reference trees. a,** Ribosomal-protein 2D tree. **b,** 16S rRNA 3D tree. Colors and hulls identify Bacteria, Archaea and Eukaryota. The highlighted internal segment marks the branch leading to Eukaryota; the black marker identifies the yeast branch.

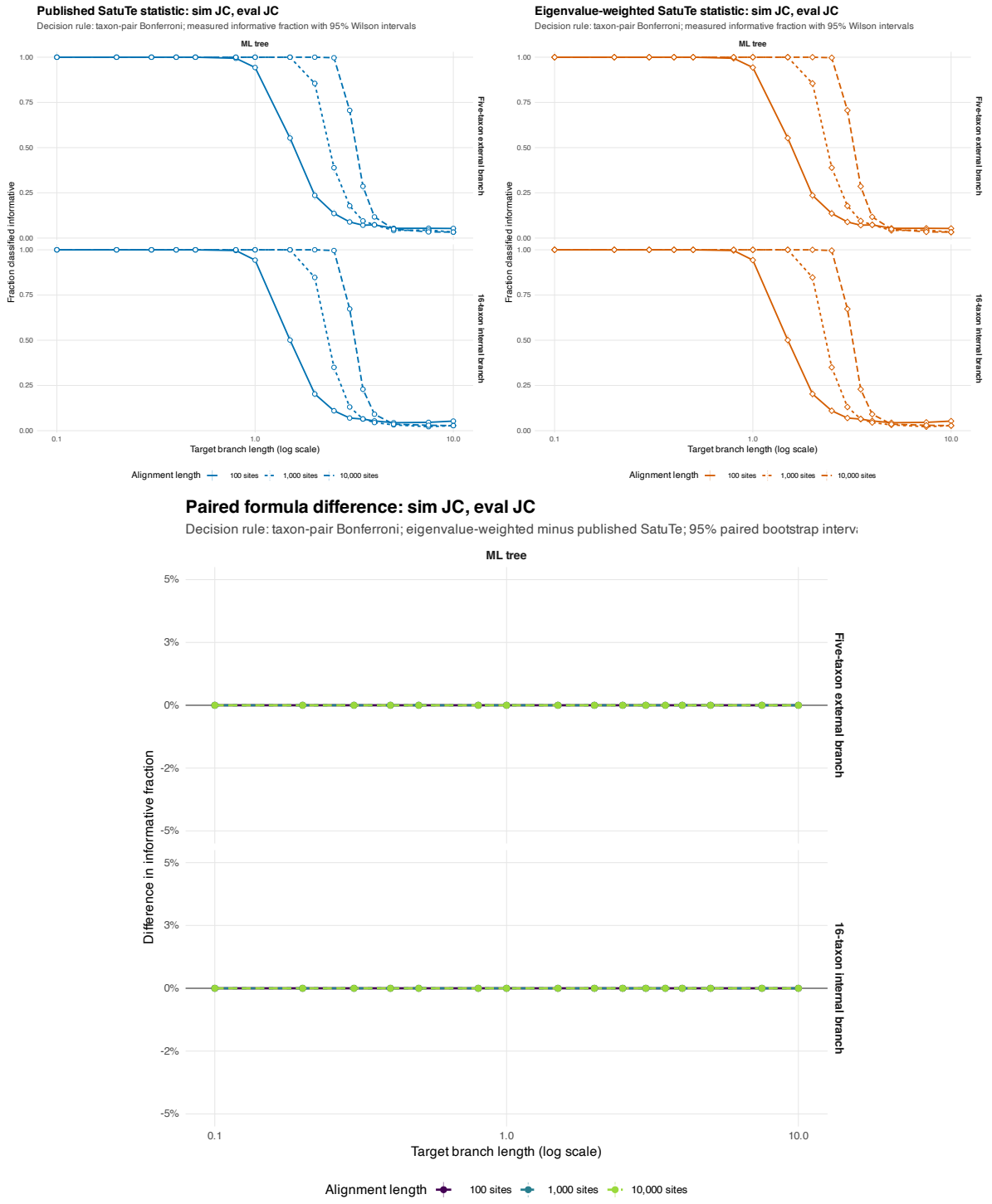


Figure S3: **JC power in the common comparison layout.** Published SatuTe (top left), eigenvalue-weighted SatuTe (top right) and their paired difference (bottom) under taxon Bonferroni. The two statistics are identical under JC, so the paired difference is exactly zero in all 288 design cells.

	Two external attachments			One external attachment			Correct topology (%)		
t	100	1,000	10,000	100	1,000	10,000	100	1,000	10,000
0.1	0	0	0	10	0	0	63.2	100.0	100.0
0.5	29	0	0	201	0	0	29.0	99.3	100.0
1.0	317	0	0	442	9	0	4.5	82.5	100.0
2.0	736	386	1	238	427	22	0.1	4.3	74.4
3.0	783	697	378	206	279	475	0.0	0.3	3.2
5.0	834	776	699	160	211	291	0.0	0.0	0.0
10.0	828	809	722	162	182	270	0.0	0.0	0.0

Table S2: **Representative split-placement results.** Counts among 1,000 inferred 16-taxon ML trees in which the branch separating the two eight-taxon subtrees attaches two external branches or one external and one internal branch, together with full-topology reconstruction accuracy. The omitted intermediate branch lengths follow the same monotone deterioration reported in the baseline experiment. Columns are alignment lengths in sites.

Focal t	Split present	Mean bootstrap	Minimum bootstrap	SatuTe informative (%)
0.1	993	93.32	14	100.0
0.2	999	99.19	74	100.0
0.5	1,000	99.99	97	100.0
0.8	1,000	100.00	100	99.1
1.0	1,000	100.00	100	92.4
1.5	1,000	100.00	100	49.7
2.0	1,000	100.00	100	21.6
2.5	1,000	100.00	100	9.6
3.0	1,000	100.00	100	5.9
3.5	1,000	100.00	100	4.3
4.0	1,000	100.00	100	4.6
5.0	1,000	100.00	100	3.8
7.5	1,000	100.00	100	3.5
10.0	1,000	100.00	100	4.3

Table S3: **Bootstrap–saturation contrast.** For 1,000 JC alignments of 100 sites on the 16-taxon tree, bootstrap support for the inferred split remains perfect at long focal branches while the Bonferroni SatuTe informative fraction approaches the 5% null level. This is the empirical basis for treating SatuTe as complementary to bootstrap support.

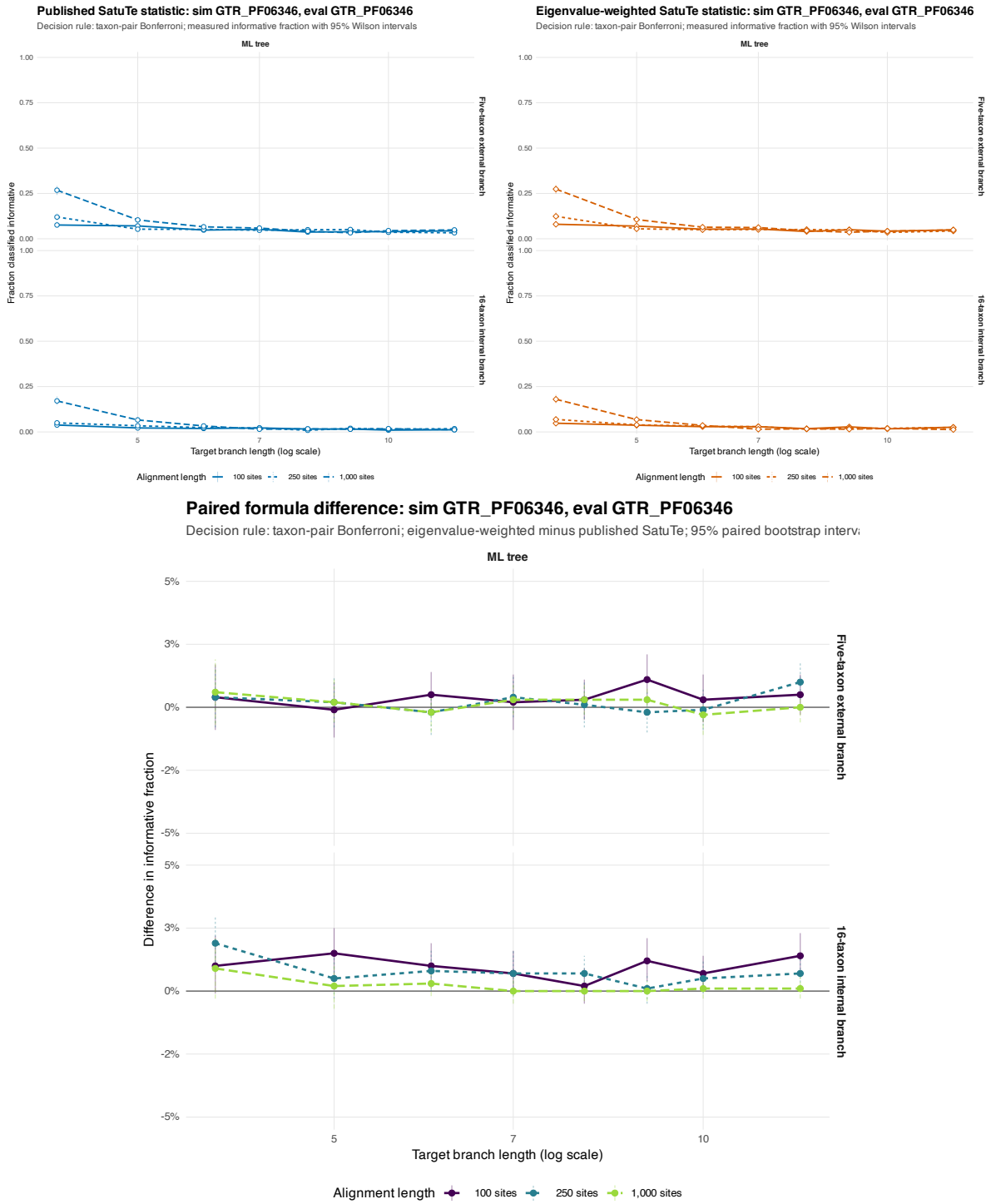


Figure S4: **Matched GTR+F power.** Published SatuTe (top left), eigenvalue-weighted SatuTe (top right) and their paired difference (bottom) under taxon Bonferroni. Differences are small and mixed in this homogeneous-rate control.

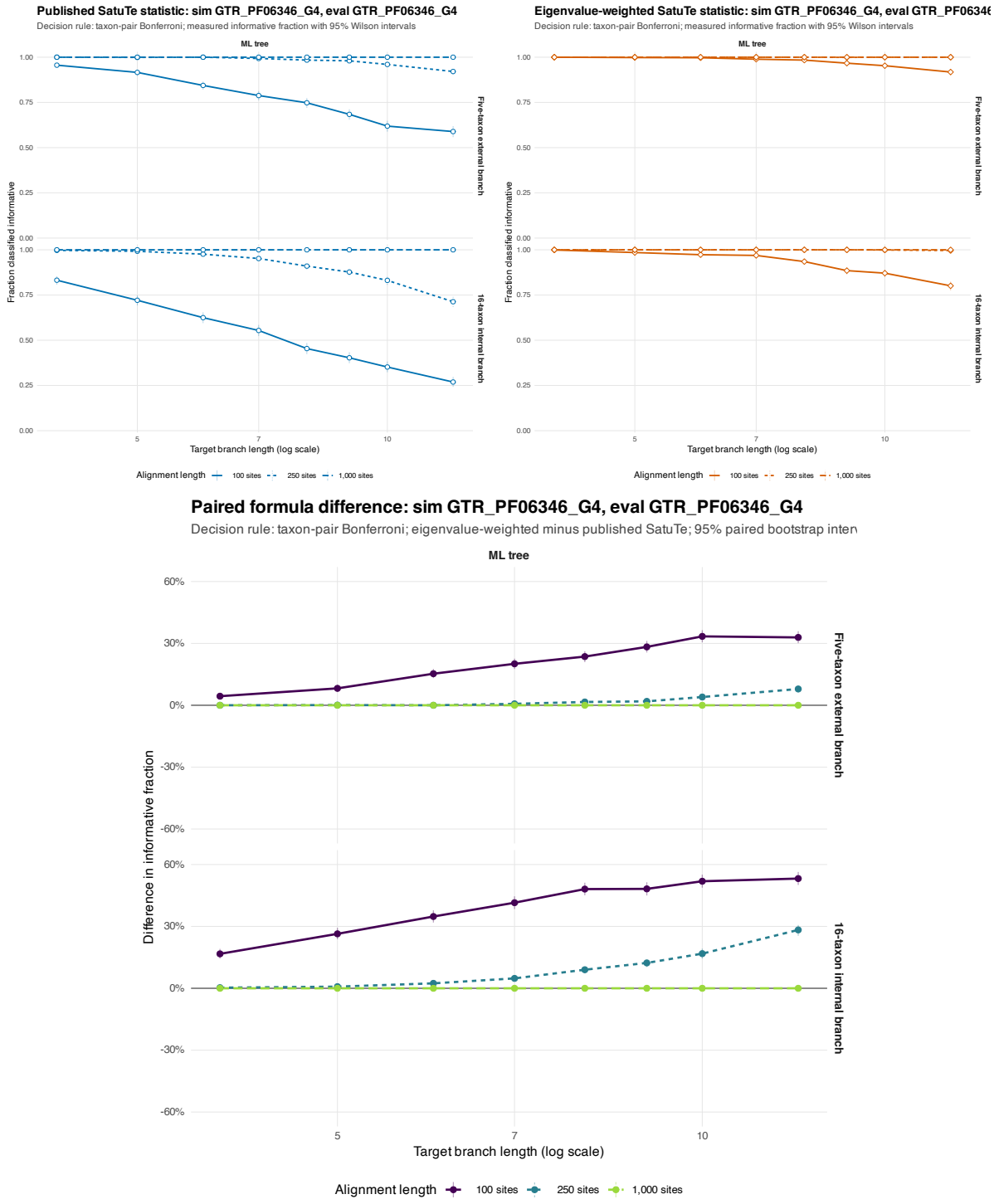


Figure S5: **Matched GTR+F+G4 power.** Published SatuTe (top left), eigenvalue-weighted SatuTe (top right) and their paired difference (bottom) under taxon Bonferroni. The eigenvalue-weighted statistic is never worse across the evaluated cells and yields gains of up to 57.6 percentage points.

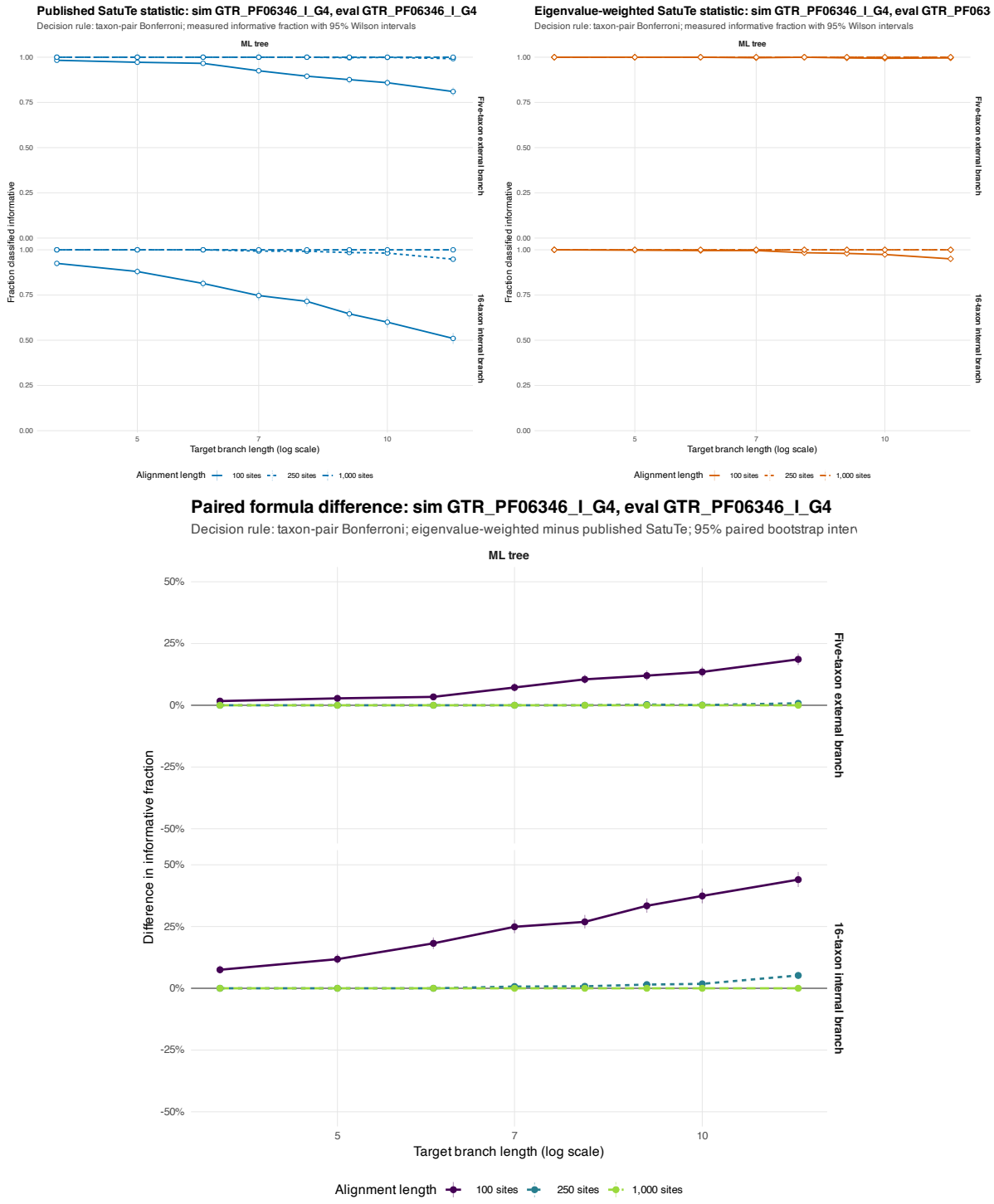


Figure S6: **Matched GTR+F+I+G4 power.** Published SatuTe (top left), eigenvalue-weighted SatuTe (top right) and their paired difference (bottom) under taxon Bonferroni. The invariant category is handled by the corrected mixture null; eigenvalue weighting is never worse across the evaluated cells and gains up to 51.1 percentage points.

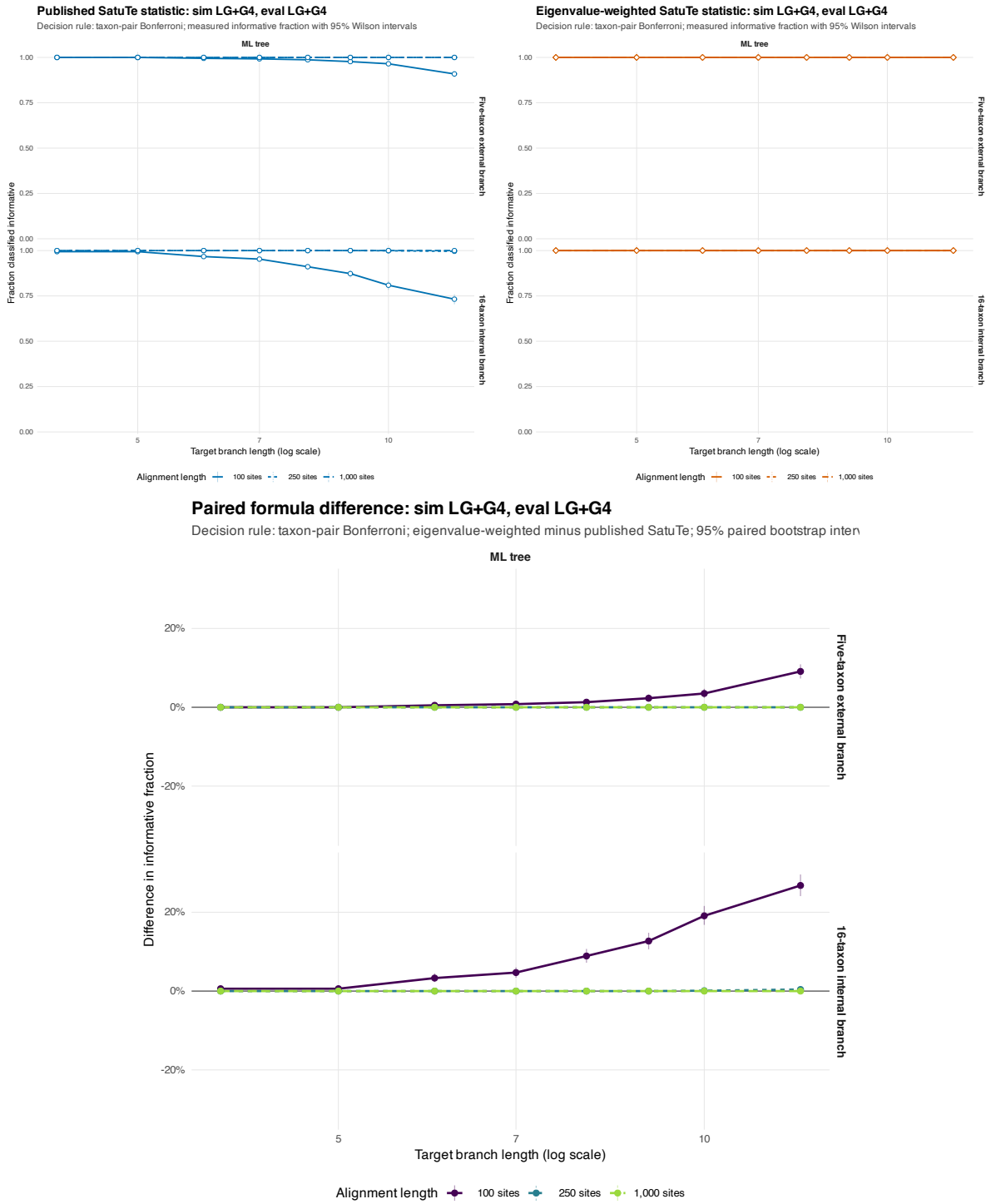


Figure S7: **Matched LG+G4 power.** Published SatuTe (top left), eigenvalue-weighted SatuTe (top right) and their paired difference (bottom) under taxon Bonferroni. Eigenvalue weighting is never worse across the evaluated cells and gains up to 32.7 percentage points.

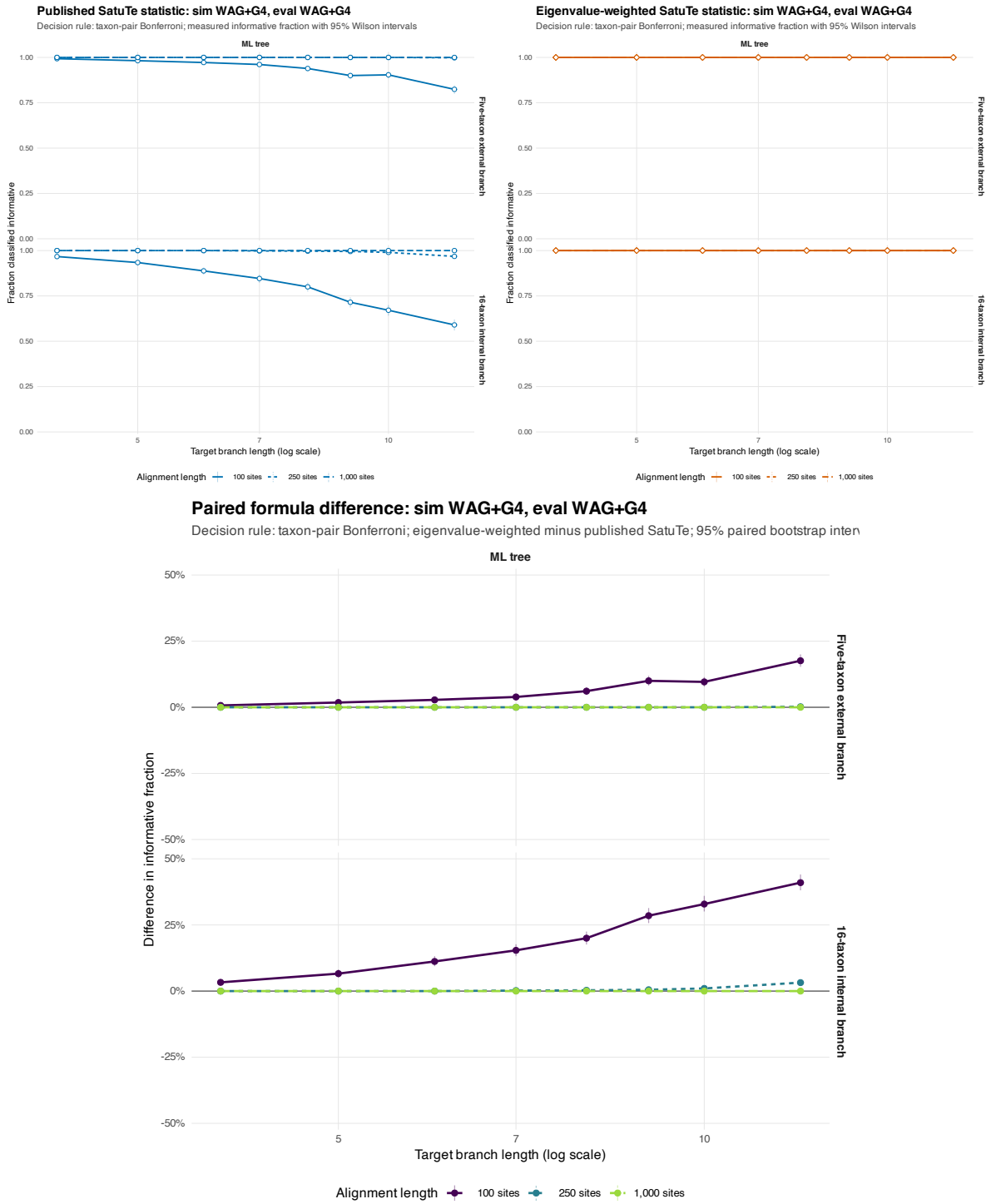


Figure S8: **Matched WAG+G4 power.** Published SatuTe (top left), eigenvalue-weighted SatuTe (top right) and their paired difference (bottom) under taxon Bonferroni. Eigenvalue weighting is never worse across the evaluated cells and gains up to 46.9 percentage points.

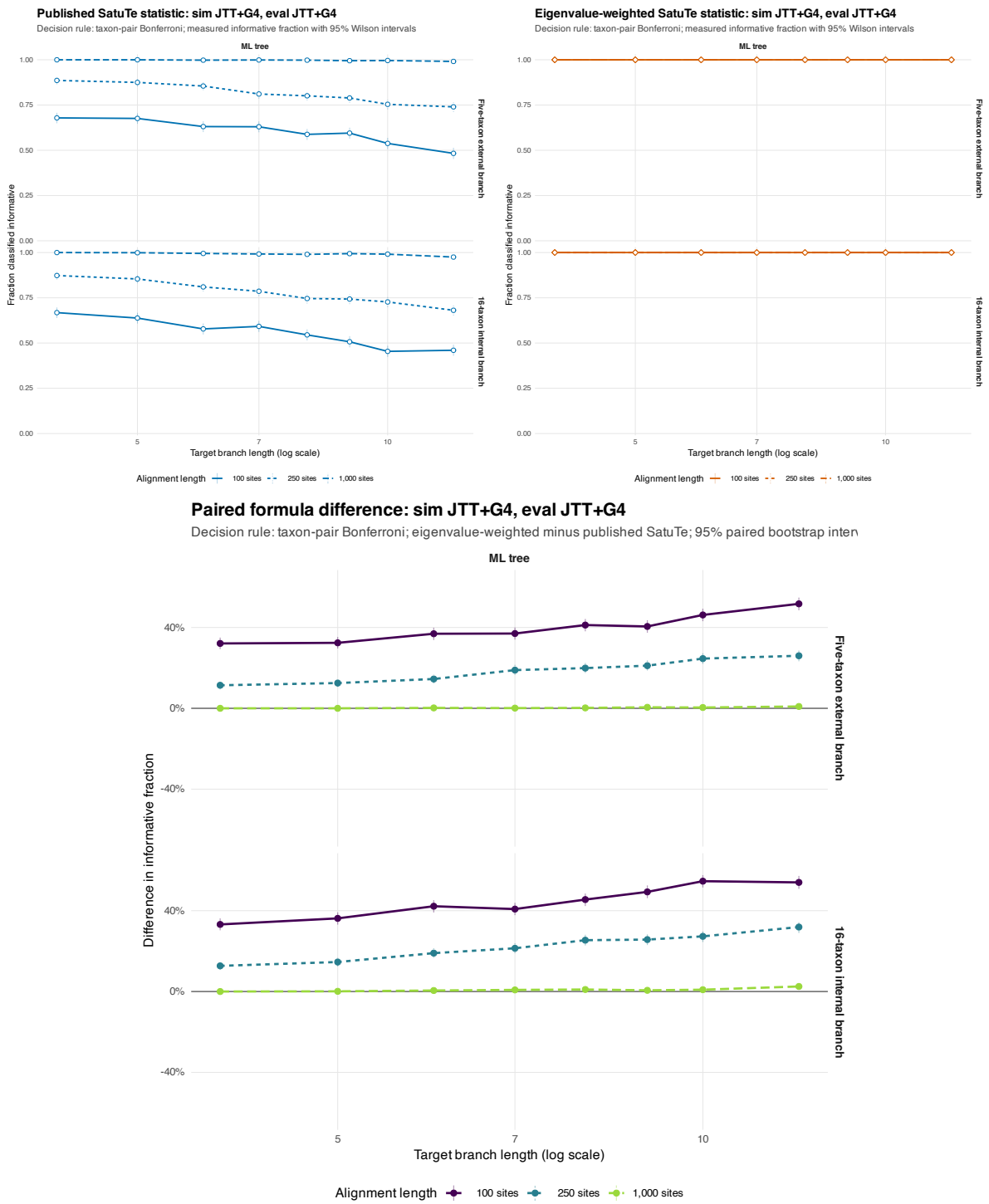


Figure S9: **Matched JTT+G4 power.** Published SatuTe (top left), eigenvalue-weighted SatuTe (top right) and their paired difference (bottom) under taxon Bonferroni. Eigenvalue weighting is never worse across the evaluated cells and gains up to 55.5 percentage points.

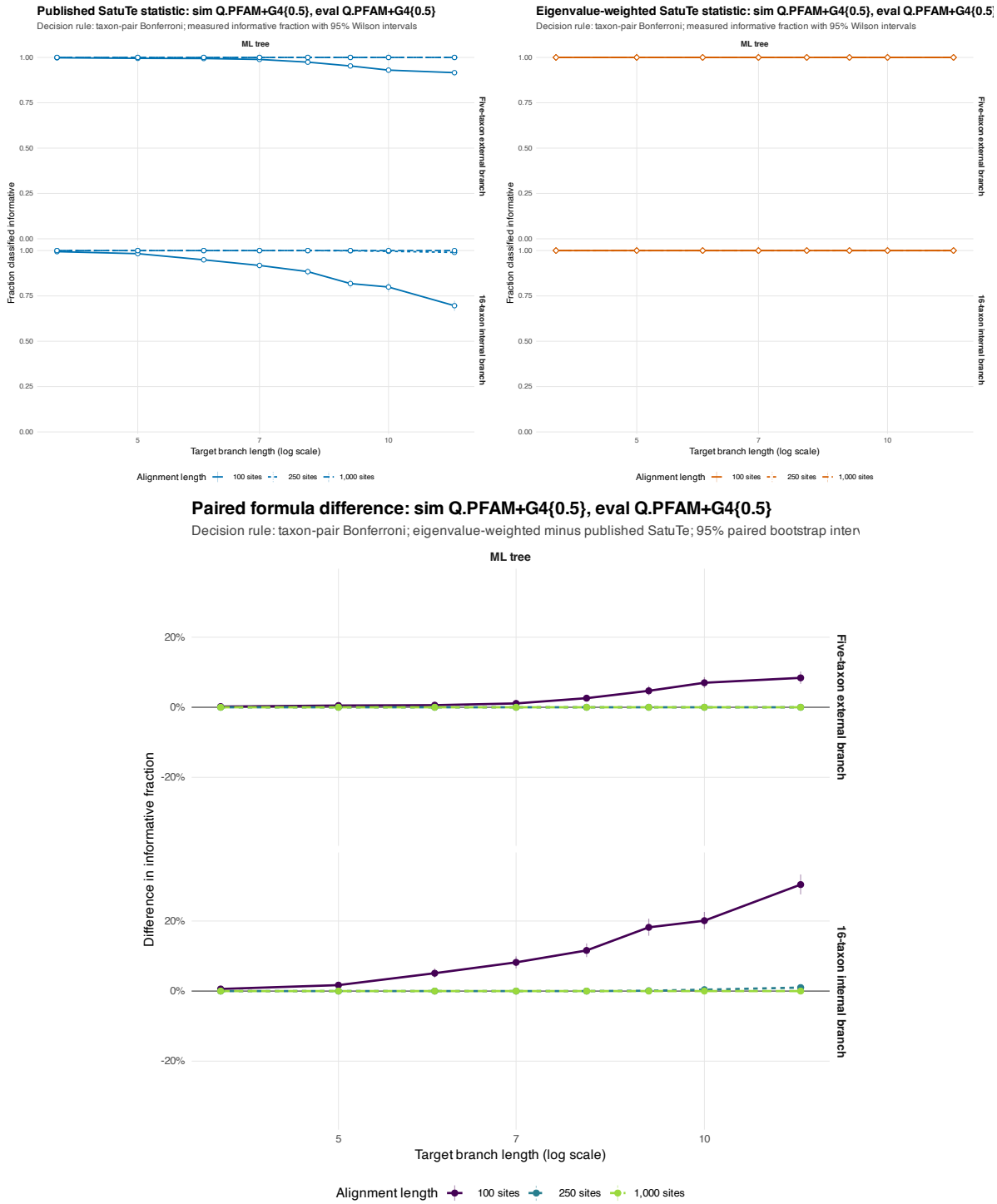


Figure S10: **Matched Q.pfam+G4 power.** Published SatuTe (top left), eigenvalue-weighted SatuTe (top right) and their paired difference (bottom) under taxon Bonferroni. Eigenvalue weighting is never worse across the evaluated cells and gains up to 35.4 percentage points.

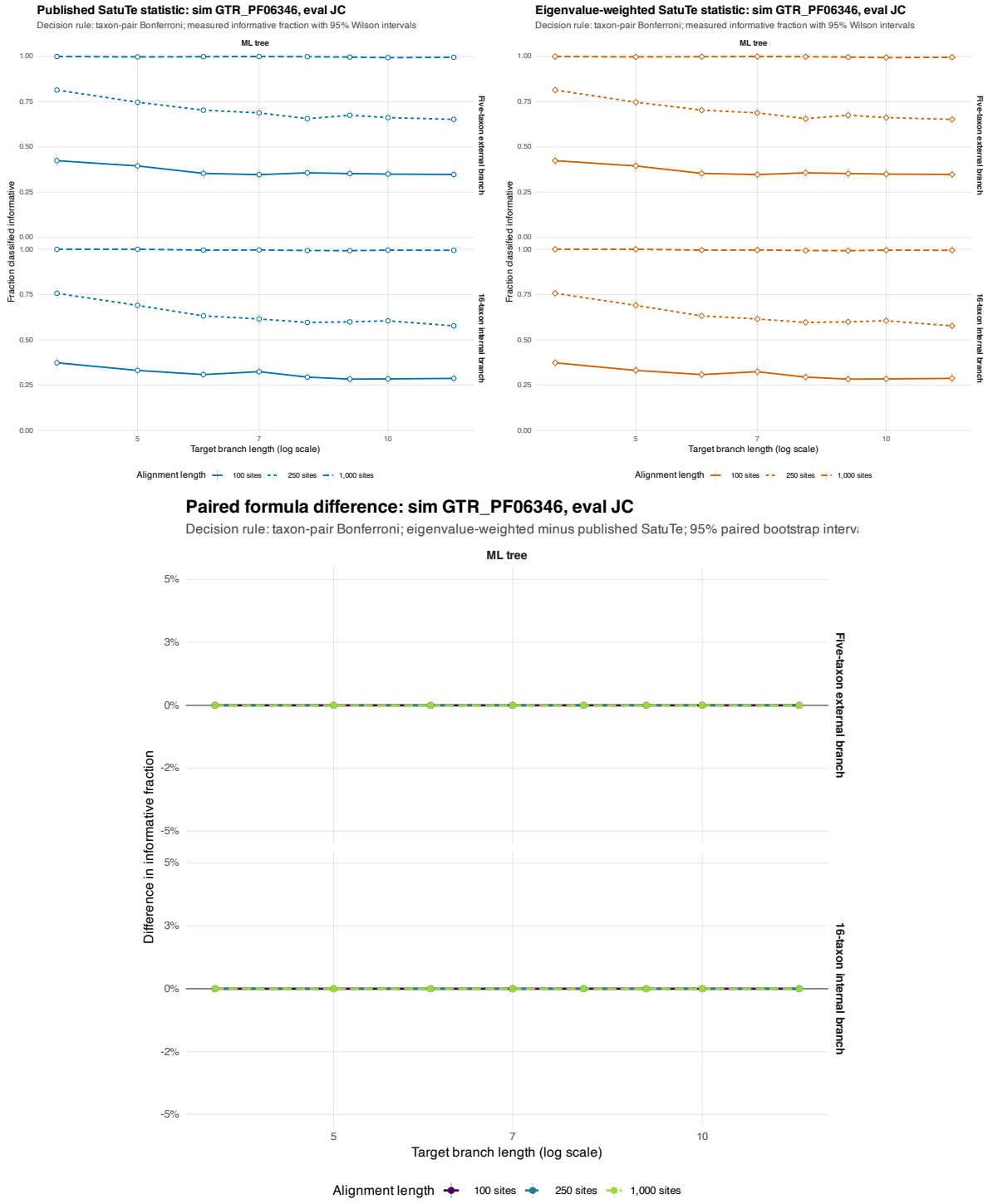


Figure S11: **GTR data evaluated by JC.** Published SatuTe (top left), eigenvalue-weighted SatuTe (top right) and their paired difference (bottom) under taxon Bonferroni. The two statistics coincide under the JC evaluation model.

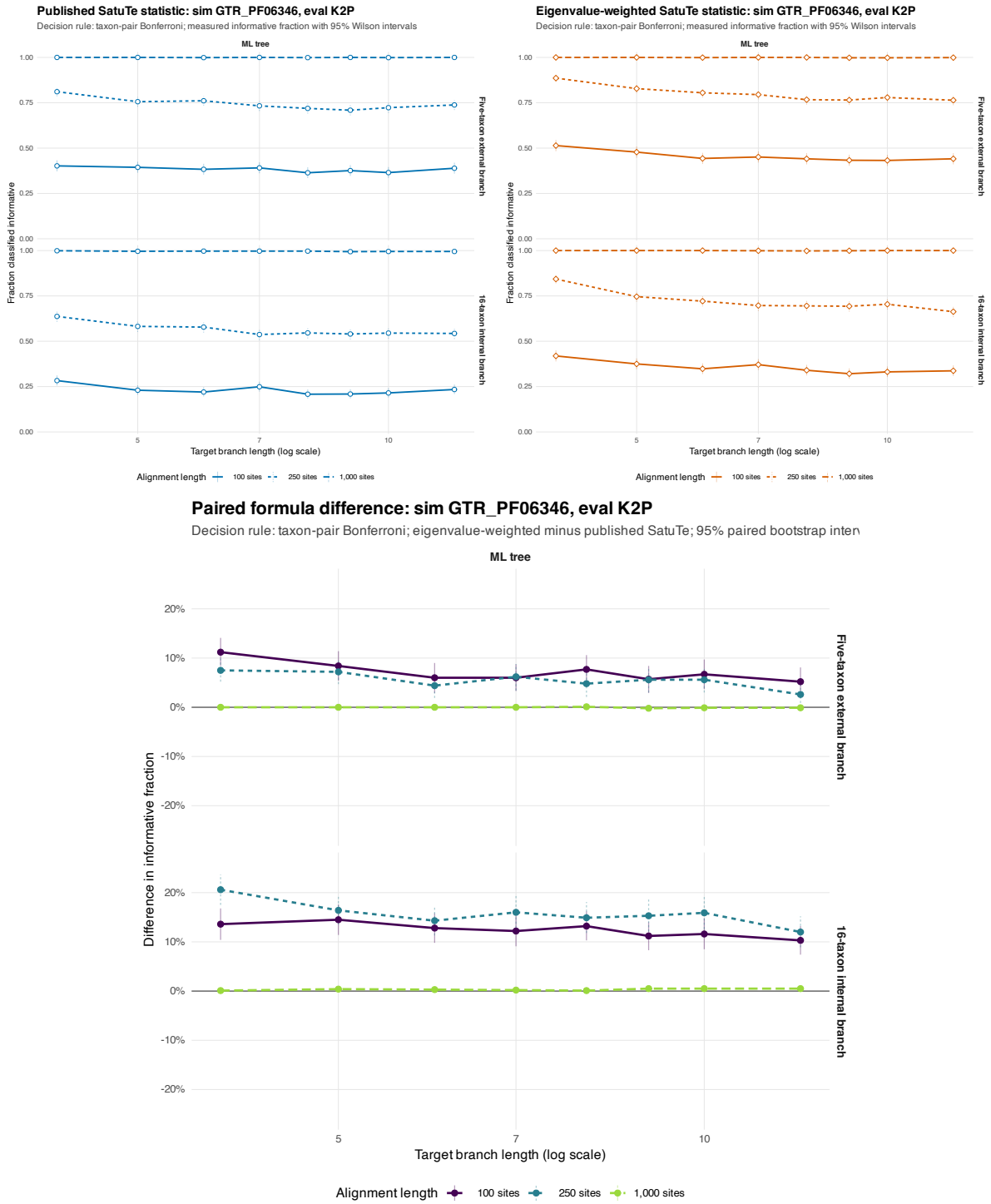


Figure S12: **GTR data evaluated by K2P.** Published SatuTe (top left), eigenvalue-weighted SatuTe (top right) and their paired difference (bottom) under taxon Bonferroni. Most cells favour eigenvalue weighting, although a minority show small negative differences.

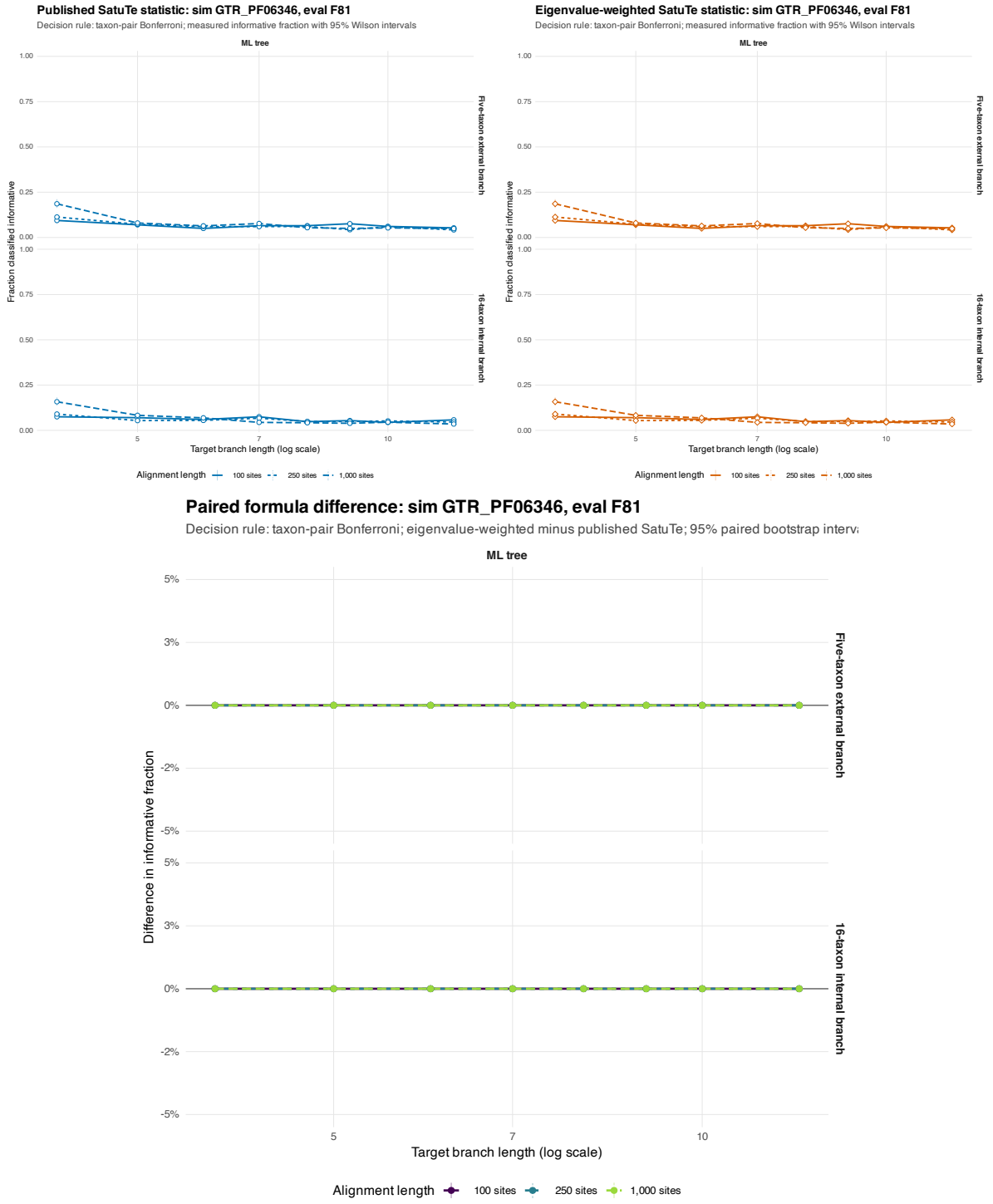


Figure S13: **GTR data evaluated by F81.** Published SatuTe (top left), eigenvalue-weighted SatuTe (top right) and their paired difference (bottom) under taxon Bonferroni. The two statistics coincide under this evaluation model.

Protein	Sites	2D	3D	ΔZ	Protein	Sites	2D	3D	ΔZ
L2	317	11.68	11.65	0.03	L18	141	8.23	8.11	0.12
L3	228	11.62	11.07	0.55	L22	174	8.28	8.69	-0.41
L4	245	11.06	11.02	0.04	L24	81	7.77	7.68	0.09
L5	164	12.31	12.21	0.10	S3	216	10.85	10.51	0.34
L6	220	10.34	10.27	0.07	S8	141	10.03	9.77	0.26
L14	125	9.96	9.01	0.95	S10	104	8.91	9.04	-0.13
L15	122	7.72	8.75	-1.03	S17	74	7.92	7.37	0.55
L16	153	6.35	6.30	0.05	S19	91	8.35	8.01	0.34

Table S4: **Per-protein topology comparison.** SatuTe Z-scores for the Eukaryota branch under the protein 2D tree and the rearranged 3D tree, with $\Delta Z = Z_{2D} - Z_{3D}$. All proteins remain significant under both placements; L15, L22 and S10 favour the rearranged 3D placement.

rRNA region	Alignment sites	3D Z	Rearranged 2D Z	$Z_{2D} - Z_{3D}$
R1	0–114	4.57	5.08	0.51
R2	115–431	6.13	5.94	-0.19
R3	432–621	3.63	3.45	-0.17
R4	622–1066	6.81	6.83	0.03
R5	1067–1221	8.99	8.93	-0.06
R6	1222–1394	8.55	8.31	-0.23
R7	1395–1565	8.93	8.35	-0.59
R8	1566–1765	12.44	12.33	-0.12
R9	1766–1946	9.33	9.65	0.33

Table S5: **rRNA-region topology comparison.** Six of nine regions have negative $Z_{2D} - Z_{3D}$ and therefore a larger score under the 3D placement.

Partition	Branch	Windows	Published SatuTe	Eigenvalue-weighted	Weighted fraction
Positions 1+2	Genotype 1	507	1	1	0.2%
	Genotype 2	507	4	1	0.2%
	Genotype 3	507	5	1	0.2%
Position 3	Genotype 1	252	11	6	2.4%
	Genotype 2	252	34	24	9.5%
	Genotype 3	252	33	27	10.7%

Table S6: **HCV saturated-window counts.** Counts use $z < z_{0.01}$ for three subtype-pair branches. Positions 1+2 and position 3 contain equal 36-site windows but span different genomic widths.